

LLM Week 5

1) What are all the challenges one would face while building a vocabulary from a large corpus of text (assume the corpus contains only an English text)?

1 point

- ☒ Deciding the size of the vocabulary
- ☒ Method to handle unknown tokens
- ☒ handling misspelled words
- ☒ handling names and numbers
- ☐ None of these

Yes, the answer is correct.

Score: 1

Accepted Answers:

Deciding the size of the vocabulary
Method to handle unknown tokens
handling misspelled words
handling names and numbers

2) Select correct statements regarding character level tokenization.

1 point

- ☐ The size of vocabulary by character level tokenization will be larger than word level tokenization, if the corpus consists of all the books published on Harry Potter.
- ☒ Computing softmax probabilities will be easy.
- ☒ The vocabulary will not expand if new words/sentences are added to the corpus.
- ☒ No issue of handling unknown tokens

3) For which of the following options, applying BPE (based on white-space delimiter as a pre-tokenization step) is *NOT* feasible

1 point

- ☒ (Japanese) 私の名前はジョン・スミスです
- ☐ (Turkish) benim adim John Smith
- ☐ (English) My name is John Smith
- ☐ (German) Mein Name ist John Smith

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Japanese) 私の名前はジョン・スミスです

white space
is no such
thing called
space character
as in
Japanese lang.

Common data for rest of the Questions below

Consider the following dictionary with the number of word occurrences in a corpus:

```
1 wo = {"low": 4,
2       "older": 5,
3       "finest": 6,
4       "lowest": 7,
5       "loneliest": 8}
```

5 { L → 4
O → 4
W → 2
D → 1
E → 5

5 { R → 1
F → 1
I → 2
N → 2
S → 3

2 { T → 3
<W> → 5

Note: Append "<w>" to each word at the end.

4) You will be learning a byte pair encoding, answer the following questions in that context:

How many tokens are there in the initial vocabulary?

12

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 12

5) Which of the following pairs has the least frequency before any merge?

1 point

- ☒ ('e', '< /w >') $w \rightarrow \text{low} \times 4 \Rightarrow 4$
- ☐ ('l', 'd') $\rightarrow \text{older} \times 5 \Rightarrow 5$
- ☐ ('l', 'n') $\rightarrow \text{finest} \times 6 \Rightarrow 6$
- ☐ ('n', 'e') $\rightarrow \text{finest} \times 6 \Rightarrow 6$
- ☐ ('w', 'e') $\rightarrow \text{lowest} \times 7 \Rightarrow 7$
- ☐ None of these.

6) What is the most frequent byte-pair before the very first merge? Say the most frequent byte pair is ('a', 'b'), then enter "ab" (without quotes and white spaces).

es

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: String) es

es
st
t </w> } all these occur
6 + 7 + 8 ⇒ 21 times

1 point

7) What is the frequency of the most frequent byte-pair before the very first merge?

21

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 21

1 point

8) How many tokens in the vocabulary will have their frequency revised/reduced after first merge?

2

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 2

only 2 tokens
e & s will get their frequency revised

1 point

9) After the first merge, how many tokens are there in the updated vocabulary?

13

12 initially + 1 (w) = 13

10) Suppose the process of building the vocab stopped after 7 merges. The following is the list of first 7 merges in that order:

1 point

Merge No.	Token 1	Token 2	Merged
1	'e'	's'	'es'
2	'es'	't'	'est'
3	'est'	'</w>'	'est</w>'
4	'l'	'o'	'lo'
5	'lo'	'w'	'low'
6	'lo'	'n'	'lon'
7	'lon'	'e'	'lone'

How will the word 'finest</w>' be tokenized?

('f', 'i', 'n', 'est</w>')

☐ ('f', 'i', 'n', 'est', '</w>')

☐ ('f', 'i', 'n', 'e', 's', 't</w>')

☐ ('f', 'i', 'n', 'e', 'st', '</w>')

☐ ('fin', 'est</w>')

☐ None of these

f i n e s t <w>
f i n est</w>

Yes, the answer is correct.

Score: 1

Accepted Answers:

You will be preparing the vocabulary with WordPiece algorithm. Answer the following questions in this context:

11) What is the highest scoring byte-pair before the very first merge? Say the highest scoring byte pair is ('a','b'), then enter "ab" (without quotes and white spaces).

fi

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: String) fi

$\frac{\text{count}(a, b)}{\text{count}(a) \text{count}(b)}$ if a and b both occur 6 & 14 times & they together occur also 6 times. which means whenever they occur, they occur together.

1 point

12) Say the score of the very first pair merged is s_1 . What is the value of $100 \cdot s_1$?

7.14

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Range) 7.09,7.19

$$\frac{6}{6 \times 14} = \frac{1}{14} \times 100 = 0.071 \\ = 7.1$$

1 point